



SKYCALC Sky Model Calculator

Sky Model Mode Version 2.0.9

Description

FAQ

Optional Parameter Initialisation using Almanac Service

Site

Observatory2640m (VLT Cerro Paranal)
Paranal -24°.6274, -70°.4039 (-24°37:38 , -70°24:14)
Apparent horizon altitude = -1°.95

Time

UT

10/12/2025 00:52:04

MJD

61019.03615741

Time step forward/back

1

hour

+

-

Paranal Local Civil Time LCT = 21:52:04
Paranal Local Sidereal Time LST = 01:26:30

First third of night

Target

Target coordinates from CDS Simbad searching by
Name or ID: hd 21520

search

α 03:26:33.685995

δ -43:36:48.838787

toggle units

Barycentric velocity correction $dv = 11.684$ km/s, $dz = 3.897e-5$

Hour Angle HA = 21:59:57

Target az = 135°.29 alt = 59°.02

Zenith distance = 30°.98

Airmass = 1.166

Ecliptic heliocentric coords = 129°.29 , -59°.17

Galactic coords = 251°.07 , -54°.76

Moon/target separation = 105°.73

Sun

Sun alt = -18°.14 , az = 233°.07

$\alpha_{\odot} = 17:08:43$ $\delta_{\odot} = -22:54:43$

Sunrise / noon / set = 09:38 / 16:32 / 23:29 UT

Solar radio flux data is not yet available

Note: The Cerro Paranal Advanced Sky Model does **not** take telluric scattered sun light into account

Moon

Moon alt = -40°.42

$\alpha_{\text{c}} = 09:59:31$, $\delta_{\text{c}} = 14:18:46$

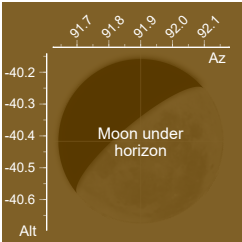
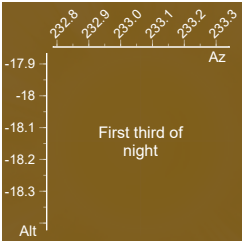
Moonrise / culm. / set = 03:47 / 09:42 / 15:35 UT

Phase angle=248°.80 , FLI $\approx$ 0.681 Waning

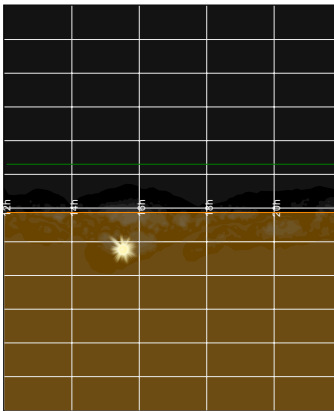
Position angle=109°.39 , Zenith angle=219°.73

Earth/Moon distance normalised = 1.00178

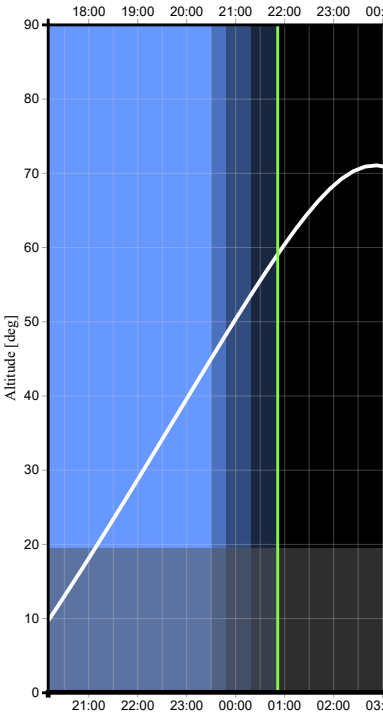
Apparent diameter = 0°.51692 = 00°31'00"



Coordinate system: Horizontal



Sun and Moon symbols are no
Visibility plot. Note that the green lin



reset almanac

transfer information to skymodel form

The Cerro Paranal Advanced Sky Model

Observatory Height above Sea Level2640m (VLT Cerro Paranal) ▾

Altitude of Target above Horizonalt=70 [19.5, 90]°
zenith distance $z=20^\circ$

*alt, z and x are coupled through the
plane parallel approximation $x=\sec(z)$,
z being the zenith distance $z=90^\circ-alt$*

Airmassx = 1.064 [1, 3]

Season and Period of NightSeason: Dec/Jan ▾
Night Period: First third of Night ▾

Precipitable Water VaporPWV = 1.0 ▾ mm

Monthly Averaged Solar Radio Flux150 ▾ sfu

Select Components to Include in the Radiance Model:

☒ Scattered Moonlight

*Note the following moon coordinate constraints: $|z - z_{\text{moon}}| \leq \rho \leq |z + z_{\text{moon}}|$
where ρ =moon/target separation, $z=90^\circ-alt$ and $z_{\text{moon}}=90^\circ-alt_{\text{moon}}$*

Sun-Moon Separation (Moon Phase)111 [0,360]° FLI ≈ 0.679 (Waxing)

Moon-Target Separation (ρ)106 [0,180]°

Moon Altitude over Horizon (alt_{moon})-28.6 [-90, 90]° $z_{\text{moon}}=118.6^\circ$

Moon-Earth Distance Normalised0.9946 [0.91, 1.08]

☒ Scattered Starlight

☒ Zodiacal Light

Heliocentric Ecliptic Longitude of Target129.19 [-180, 180]°

Ecliptic Latitude of Target-59.17055 [-90, 90]°

☒ Molecular Emission of Lower Atmosphere

☒ Emission Lines of Upper Atmosphere

☒ Airglow/Residual Continuum

☒ Instrumental Thermal Emission

This radiance component represents an instrumental effect. The emission is provided relative to the other model components. To obtain the correct absolute flux, an instrumental response curve must be applied to the resulting model spectrum (see [section 6.2.4](#) in the [documentation](#)).

Element₁ (e.g. Telescope) T₁ = 275 K Emissivity₁ = 0.25 [0,1[

Element₂ (e.g. Instrument) T₂ = 70 K Emissivity₂ = 0.27 [0,1[

Element₃ (e.g. Cryostat) T₃ = 70 K Emissivity₃ = 0.00 [0,1[

Wavelength Grid

☐ Vacuum

☒ Air (using the formula of [Edlén 1966](#))

λ_{min} = 1050 nm [≥ 300 nm]

λ_{max} = 1850 nm [$\leq 30,000$ nm]

☐ Linear binning $\Delta\lambda$ = nm/bin

Convolving Line Spread Function

☐ None

☒ Gaussian FWHM=2 bins

☐ Boxcar Width= bins

☒ Logarithmic binning $\lambda/\Delta\lambda =$ [$\leq 10^6$]

Bins: 169919 bins
 $\lambda/\Delta\lambda$: 300000 (max 1000000)
 FITS file size: 9.4Mb (max 30 Mb)
 Approx. response time: 5.2 s

Outputs

- ☒ Plot Radiance Spectrum ☒ Plot Radiance Components
- ☒ Plot Transmission Spectrum ☒ Plot Transmission Components
- ☐ Calculate Broad-Band UBVRIZYJHKLMNQ magnitudes per arcsec²
- ☐ Plot LSF Kernel (select an LSF kernel to enable this option)

Submit

Reset

SkyCalc is based on the [Cerro Paranal Advanced Sky Model](#), which was developed in particular to be used in the [ESO Exposure Time Calculators](#), by a team of astronomers at the Institute for Astro- and Particle Physics at the University of Innsbruck, as part of an Austrian in-kind contribution to ESO.

The in-kind contribution also includes two tools to correct observations for telluric absorption and emission (Molecfit and Skycorr), which can be found [here](#).

The citations for the Cerro Paranal Sky Model are [Noll et al. \(2012, A&A 543, A92\)](#) and [Jones et al. \(2013, A&A 560, A91\)](#).

[A library of telluric transmission spectra](#) with various resolutions and atmospheric conditions has been calculated with the Cerro Paranal Advanced Sky Model. A description of the usage of the spectra can be found in the article [Moehler et al. \(2014, A&A 568, A9\)](#).

Send comments and questions via <https://support.eso.org/>